

Beyond 1.58 Bits: Adaptive-Precision Ternary Transformers with Fully-Ternary Mixture-of-Experts Routing

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Abstract

BitNet established that large language models can be trained with ternary weights $\{-1, 0, 1\}$ —effectively 1.58 bits per parameter—without loss of modeling power at scale. We argue that 1.58 bits is a design point, not a floor, and present a unified framework that pushes ternary architectures in three directions at once. **(1) Below the floor:** we give an analytic comparison of binary $\{-1, +1\}$ versus ternary quantization under a Laplace weight prior, proving that optimal-threshold ternary quantization reduces expected squared quantization error to $2(1 - e^{-1/2}) \approx 0.787$ of the binary optimum, and we characterize exactly when the zero state earns its 0.58 bits. **(2) Around the floor:** we formulate heterogeneous precision as a constrained bit-budget optimization, allocating per-module precision by sensitivity rather than imposing uniform 1.58 bits everywhere. **(3) Above the floor:** we describe a fully-ternary mixture-of-experts architecture in which *all* experts—not merely the routed ones—are ternary, the precision budget instead being spent on the router, whose parameters are few but high-leverage. The three contributions are instances of a single Lagrangian bit-budget framework. We relate our design to concurrent ternary-MoE efforts (MoTE, TC-MoE), from which it differs by removing the full-precision shared expert entirely, and we state our empirical boundaries honestly: the analytic results are proved, the implementation exists, and full-scale training validation is ongoing work.

1 Introduction

The BitNet line of work [1, 2] demonstrated something the field had assumed impossible: Transformers trained natively with weights constrained to $\{-1, 0, +1\}$ match full-precision models in perplexity and downstream performance once the parameter count crosses a few billion. The headline number— $1.58 = \log_2 3$ bits per weight—became both a result and, implicitly, a wall. Subsequent engineering effort went into serving (bitnet.cpp), activation quantization (BitNet a4.8), and adaptation (LoTA-QAF), but the weight precision itself was treated as settled.

We take the opposite view: 1.58 bits is one point in a much larger design space, and the interesting questions all lie *around* it rather than *at* it:

- **Below:** When is 1 bit enough, and what exactly does the ternary zero buy over binary? The answer turns out to be analytic, not empirical.
- **Around:** Why should every layer pay 1.58 bits? Sensitivity varies by orders of magnitude across modules; precision should be budgeted, not uniform.
- **Above:** In a mixture-of-experts, where should the precision budget go—the many expert parameters, or the few router parameters that decide which experts fire?

This paper makes three contributions, unified by a single constrained optimization framework over per-module bit budgets (Section 4):

1. An **analytic sub-1.58-bit theory** (Section 5): under an i.i.d. Laplace weight prior we derive the optimal ternary threshold $\Delta^* = b/2$, prove the ternary-to-binary mean-squared-error ratio $2(1 - e^{-1/2}) \approx 0.787$, and compute the operating-point entropy (≈ 1.574 bits of the 1.585 maximum), showing the zero state is nearly maximally informative at the optimum.
2. A **heterogeneous-precision allocation framework** (Section 6): per-module precision b_ℓ chosen by Lagrangian bit-budget optimization against measured sensitivity, with a greedy marginal-gain algorithm.
3. A **fully-ternary mixture-of-experts architecture** (Section 7): 34 ternary experts with dual routing modes (dense pull-weighted deliberation and Gumbel top- k sparse dispatch), a higher-precision router justified by leverage analysis, and precision-preserving low-rank adapters that make the design mixed-precision *by construction*.

We position against the concurrent ternary-MoE literature honestly. MoTE [3] combines a *frozen full-precision shared expert* with ternary routed experts; TC-MoE applies ternarization to routing *decisions* rather than weights. Our design removes the full-precision crutch entirely. What we do *not* claim is also stated plainly in Section 10: the theorems are proved, the modules are implemented, and large-scale training runs are in progress—not complete.

2 Related Work

1-bit and ternary LLMs. BitNet [1] introduced the BitLinear layer, replacing matmul with ternary accumulation trainable via the straight-through estimator. BitNet b1.58 [2] showed ternary weights match FP16 at 3B+ parameters and proposed a new 1-bit scaling law. FBI-LLM and subsequent reproductions confirmed the recipe; bitnet.cpp brought CPU inference; BitNet a4.8 attacked the activation side with 4-bit activations. Our work accepts this foundation and asks what lies past it.

Ternary mixture-of-experts. MoTE [3] (arXiv:2506.14435) pairs a frozen BF16 shared expert with ternary routed feed-forward experts under top-1 routing, reporting 62% expert memory reduction. TC-MoE [4] (ICLR 2025) ternarizes the routing *decisions*—positive, deactivated, negative expert combination—to combat expert underutilization. Both keep a full-precision escape hatch (shared expert) or leave weights untouched. We close that hatch: *every* expert is ternary, and the precision budget moves to the router.

Ternary adaptation. LoTA-QAF demonstrated lossless ternary adaptation: ternary adapters merge exactly into the ternary grid, avoiding the dtype-mismatch tax of FP16 LoRA [10] on quantized bases. Our rank-8 swarm adapters (Section 7) follow the same principle from the opposite direction—higher-precision adapters over a ternary base, i.e., mixed precision by construction rather than by accident.

Mixed and adaptive precision. Post-training quantization (GPTQ [5], AWQ [6], SmoothQuant) established that sensitivity is highly non-uniform: $\sim 1\%$ of weights (outliers, salient channels) dominate the error. We lift this observation from post-hoc quantization into *native* training-time architecture via the bit-budget framework of Section 4.

Sparse expert models. Switch Transformer [7], GShard, and Mixtral [8] established top- k expert routing; Gumbel-Softmax [9] gave differentiable discrete routing. Our council uses both dense pull-weighted and Gumbel top- k routing over ternary experts.

3 Preliminaries: Ternary Computation

A BitLinear layer constrains weights to $W \in \{-1, 0, +1\}^{m \times n}$ with per-tensor absmean scaling:

$$\alpha = \frac{1}{mn} \sum_{ij} |W_{ij}|, \quad W_q = \text{clip} \left(\text{round} \left(\frac{W}{\alpha} \right), -1, 1 \right). \quad (1)$$

Activations are quantized to INT8. The forward pass becomes pure addition/subtraction:

$$Y = X_{\text{int8}} \cdot W_q = \sum_{j: W_{ij}=1} X_j - \sum_{k: W_{ik}=-1} X_k, \quad (2)$$

eliminating floating-point multiplication in the dense core. Gradients flow through the non-differentiable quantizer via the straight-through estimator (STE). Analytically, 1.58-bit weights yield an 87.5% memory reduction versus FP16, and BitNet literature reports up to $4.1 \times$ operational energy efficiency over FP16 matmul; we treat these as the published analytic bounds they are, not as measurements of our system.

The alphabet size 3 gives the headline figure: maximum entropy per weight $\log_2 3 \approx 1.585$ bits. Everything that follows is about spending those bits—and fewer, and more—where they matter.

4 The Heterogeneous Bit-Budget Framework

Let a model decompose into modules $\ell = 1, \dots, L$ (layers, experts, routers, embeddings), module ℓ holding P_ℓ parameters at precision b_ℓ bits. The total bit budget is

$$B = \sum_{\ell=1}^L P_\ell b_\ell. \quad (3)$$

Uniform BitNet is the special case $b_\ell \equiv 1.58$. We instead solve

$$\min_{b_1, \dots, b_L} \sum_{\ell=1}^L S_\ell \cdot \varepsilon(b_\ell) \quad \text{s.t.} \quad \sum_{\ell=1}^L P_\ell b_\ell \leq B, \quad (4)$$

where $S_\ell \geq 0$ is the *sensitivity* of the training objective to quantization noise in module ℓ (operationally: diagonal Fisher trace, Hessian trace estimate, or ablation-measured degradation), and $\varepsilon(b)$ is the expected quantization error at b bits, assumed convex decreasing in b .

Lemma 1 (Water-filling condition). *At an interior optimum of (4),*

$$S_\ell P_\ell (-\varepsilon'(b_\ell)) = \lambda \quad \forall \ell, \quad (5)$$

for a single shadow price $\lambda \geq 0$: marginal error reduction per marginal bit is equalized across modules.

The three contributions of this paper are three instantiations of this one equation: **(1)** asks what $\varepsilon(b)$ looks like for $b \in \{1, 1.58\}$, analytically; **(2)** solves (4) across modules with measured S_ℓ ; **(3)** observes that in an MoE, $S_{\text{router}}/P_{\text{router}} \gg S_{\text{expert}}/P_{\text{expert}}$, so the budget flows to the router.

5 Below the Floor: Sub-1.58-Bit Analysis

BitNet settled an empirical question (can 1.58 bits work?) but left the analytic one open: *what does the zero buy?* We answer it exactly under a Laplace weight prior—the maximum-entropy symmetric distribution for fixed mean absolute deviation, and a good model of the heavy-tailed, zero-concentrated weight histograms observed in trained Transformers.

Definition 1 (Quantizers). *Let w have symmetric density $p(w)$ with $\mathbb{E}[w] = 0$. The ternary quantizer with threshold $\Delta \geq 0$ and scale α is $\hat{w}_T = \alpha \cdot \text{sign}(w) \mathbf{1}\{|w| > \Delta\}$; the binary quantizer is $\hat{w}_B = \alpha_B \cdot \text{sign}(w)$.*

For fixed α , mean-squared error decomposes as $\mathbb{E}[(w - \hat{w})^2] = \mathbb{E}[w^2] - 2\alpha \mathbb{E}[wQ(w)] + \alpha^2 \mathbb{E}[Q(w)^2]$. The error-minimizing scale is $\alpha^* = \mathbb{E}|w|$ (absmean—exactly the BitNet scaling), giving for binary:

$$\text{MSE}_B = \mathbb{E}[w^2] - (\mathbb{E}|w|)^2 = \text{Var}(|w|). \quad (6)$$

Theorem 1 (Ternary advantage under a Laplace prior). *Let $w \sim \text{Laplace}(0, b)$, i.e. $p(w) = \frac{1}{2b}e^{-|w|/b}$. With absmean scaling $\alpha = \mathbb{E}|w| = b$, the ternary quantizer minimizes MSE at threshold $\Delta^* = b/2$, achieving*

$$\frac{\text{MSE}_T(\Delta^*)}{\text{MSE}_B} = 2(1 - e^{-1/2}) \approx 0.787. \quad (7)$$

That is, the zero state reduces quantization error by $\approx 21\%$ relative to binary at the optimum, using at most 1.585 bits per weight.

Proof sketch. $\mathbb{E}[w^2] = 2b^2$, $\mathbb{E}[Q_T^2] = P(|w| > \Delta) = e^{-\Delta/b}$, and $\mathbb{E}[wQ_T(w)] = \mathbb{E}[|w|\mathbf{1}\{|w| > \Delta\}] = (\Delta + b)e^{-\Delta/b}$ by direct integration. Hence

$$\text{MSE}_T(\Delta) = 2b^2 - e^{-\Delta/b}(2b\Delta + b^2).$$

Differentiating, $\frac{d}{d\Delta}\text{MSE}_T = -e^{-\Delta/b}(2\Delta - b)$, so the unique stationary point is $\Delta^* = b/2$, a minimum since the derivative is negative at $\Delta = 0$ (ternary strictly beats binary for any small $\Delta > 0$). Substituting, $\text{MSE}_T(b/2) = 2b^2(1 - e^{-1/2})$, while $\text{MSE}_B = b^2$. $\square$

Remark 1 (When the zero earns its keep). *The proof shows the operative condition is distributional, not architectural: ternary beats binary iff the weight density concentrates enough mass near zero for the dead-zone to absorb variance rather than signal—precisely the leptokurtic (heavy-tailed, zero-spiked) regime of trained LLM weights. For platykurtic (uniform-like) weights the advantage vanishes, and binary is the correct budget choice. This is a per-tensor decision rule, and it is the analytic core of the adaptive framework in Section 6.*

At the optimum, the zero-state occupancy is $p_0 = P(|w| \leq b/2) = 1 - e^{-1/2} \approx 0.393$, and the operating-point entropy is

$$H = -p_0 \log_2 p_0 - (1 - p_0) \log_2 \frac{1 - p_0}{2} \approx 1.574 \text{ bits}, \quad (8)$$

versus the $\log_2 3 \approx 1.585$ maximum: the zero state is *nearly maximally informative* at the MSE optimum. The 0.58 bits are not overhead—at the optimal operating point, they are almost fully used.

Design consequence. The analysis yields a concrete mixed sub-ternary recipe: tensors whose empirical kurtosis exceeds the Laplace value ($\kappa > 6$) stay ternary; tensors near $\kappa \approx 1.8$ (uniform-like, e.g. some embeddings) drop to binary $\{-1, +1\}$ at 1.0 bits with negligible excess error. This is “below BitNet” made operational.

6 Around the Floor: Adaptive Precision Allocation

Theorem 1 tells us *which* tensors deserve the zero state. The budget framework tells us *how many bits* each module deserves overall. We now make (4) operational.

Sensitivity measurement. For each module ℓ we estimate S_ℓ by the diagonal Fisher approximation on a calibration batch, $S_\ell \approx \text{tr}(\hat{F}_\ell)$, cross-checked against single-module ablation (quantize module ℓ alone, measure validation degradation); where the two estimates disagree in rank order, the ablation measurement governs.

Error model. We fit $\varepsilon(b)$ per module class from the quantizer family: $\varepsilon(1)$ and $\varepsilon(1.58)$ from the binary/ternary analysis above, $\varepsilon(4), \varepsilon(8)$ from standard uniform-quantizer $\Delta^2/12$ noise models. Monotone convex interpolation completes the curve.

Allocation algorithm. With discrete precision levels $\mathcal{B} = \{1, 1.58, 4, 8\}$, the Lagrangian condition becomes a greedy marginal-gain procedure: start all modules at 1 bit; repeatedly spend the next bit-chunk on the module maximizing $S_\ell \cdot [\varepsilon(b_\ell) - \varepsilon(b_\ell^+)] / [P_\ell \cdot (b_\ell^+ - b_\ell)]$ until the budget B binds. This is the classic water-filling discretization and runs in $O(L|\mathcal{B}|\log(L|\mathcal{B}|))$.

The qualitative outcome is stable across calibration runs and matches the post-training-quantization literature’s lesson [6]: attention out-projections and the router are precision-hungry; the bulk feed-forward mass is ternary-robust; a minority of uniform-like tensors are binary-safe. Uniform 1.58 bits is therefore simultaneously *too much* (for binary-safe tensors) and *too little* (for the router)—the precise inefficiency the budget framework removes.

7 Above the Floor: Fully-Ternary Mixture-of-Experts

We now apply the framework where the leverage asymmetry is most extreme: sparse expert models. Our reference architecture is a 34-expert council in which *every* expert feed-forward block is ternary BitLinear, trained natively with STE. Two routing modes are supported:

- **Dense pull-weighted deliberation:** all 34 experts evaluate every token; outputs combine by learned pull weights $y = \sum_{i=1}^{34} \pi_i(x) E_i(x)$, $\sum_i \pi_i = 1$. Maximum capacity, maximum cost—used for hard reasoning steps.
- **Gumbel top- k sparse dispatch:** $k = 4$ experts per token ($\approx 11.8\%$ active parameter ratio) via Gumbel-Softmax routing with temperature annealing, for throughput-bound decoding.

Where the bits go: router leverage. The router holds $P_r = d \times 34$ parameters against experts holding $P_e \gg P_r$. But a router quantization error does not merely add noise—it *misroutes*, replacing an expert’s output with a different expert’s, an error amplified by inter-expert output distance. Formally, if router noise flips the top-1 choice with probability δ , the excess risk scales as $\delta \cdot \mathbb{E}_{i \neq j} \|E_i(x) - E_j(x)\|^2$, which dominates the $\varepsilon(1.58)$ expert weight noise whenever experts are diverse—which is the entire point of a council. The budget equation (4) with $S_{\text{router}}/P_{\text{router}} \gg S_{\text{expert}}/P_{\text{expert}}$ therefore allocates the router 8 bits (or FP16) while experts remain at 1.58. This is the architectural difference from MoTE [3]: rather than a frozen full-precision *shared expert* as a safety net, we spend the precision budget on the *decision of which expert fires*. No full-precision expert exists anywhere in the model.

Precision-preserving adapters. Each expert carries rank-8 low-rank adapter deltas (the EGGROLL swarm substrate) maintained at higher precision. Since adapters compose additively with the ternary base, the system is mixed-precision *by construction*: the frozen ternary core supplies capacity-per-bit, the adapters supply task-specific correction where the budget framework says precision is warranted. This mirrors the LoTA-QAF lesson—keep the adaptation grid compatible with the base grid—from the opposite side.

Optimization for ternary experts. Ternary STE training is stabilized with (i) SubLN normalization around each BitLinear block, bounding activation variance before quantization, and (ii) a Muon-style orthogonalized optimizer (Newton-Schulz polar decomposition of momentum), which constrains the spectral norm of updates and prevents the gradient-divergence pathology that naive Adam exhibits through STE layers. Both mechanisms are implemented in the reference codebase; neither is claimed as novel, and both are ablatable.

Remark 2 (What is actually new). *To keep the novelty claim exact: native ternary training is BitNet’s; ternary routed experts with a full-precision shared expert is MoTE’s; ternary routing decisions are TC-MoE’s [4]. Ours is the combination under one budget: the sub-1.58-bit analytic rule (Theorem 1), the heterogeneous allocation (Section 6), and the fully-ternary council with a high-precision router and no full-precision expert anywhere. Each part is independently citable; the framework is what makes them one paper rather than three.*

8 Implementation Notes

The reference implementation realizes every mechanism above as executable modules (BitLinear with absmean scaling and STE; Gumbel top- k and dense pull-weighted routers; SubLN; orthogonalized optimizer). Two properties are worth recording for reproducibility:

- **Exact ternary matmul.** The forward pass uses integer accumulation over $\{-1, 0, 1\}$ (Section 3); no floating-point multiplication occurs in expert feed-forward blocks.
- **Deterministic resource accounting.** A hardware governor measures per-step latency and throttles diffusion-refinement depth, keeping consumer-hardware inference inside a fixed energy envelope. The governor is engineering, not science, and is documented for completeness.

On a formal-logic curriculum probe, the ternary STE stack converges stably (loss 0.0789 at step 2,500), confirming gradient flow through the quantizer; full training curves are reported in the companion Quillan-Ronin technical report and are not re-claimed here.

9 Analytic Results and Bounds

Quantity	Uniform 1.58-bit	Heterogeneous (this work)
Weight memory vs. FP16	−87.5%	−87.5% to −93.75% [†]
Router precision	1.58 bit	8 bit (budgeted)
Binary-safe tensors	1.58 bit (waste)	1.0 bit
Expert matmul	adds/subs	adds/subs (unchanged)

Table 1: Analytic comparison. [†]Upper bound assumes the binary-safe tensor fraction characterized in Section 5; realized savings depend on measured kurtosis profiles.

Table 1 summarizes the analytic bounds. The headline: heterogeneous budgeting cuts an additional slice off the already-small ternary footprint *while spending more* on the router—the one place the analysis says precision has leverage. Energy follows the BitNet literature’s reported $4.1\times$ matmul efficiency for the ternary core; the router’s 8-bit matmul is negligible in the total ($P_r/P_e \ll 1$).

10 Limitations and Honest Boundaries

1. The Laplace analysis (Theorem 1) is exact for the stated prior; real weight tensors are only approximately Laplacian. The kurtosis decision rule is the bridge, and its calibration across model families is ongoing.
2. Sensitivity estimates (Fisher trace) are approximations; the allocation is only as good as S_ℓ .
3. Large-scale pre-training of the full 34-expert ternary council is in progress. We claim the framework, the theorem, the architecture, and the implementation—not a trained artifact’s benchmark scores.
4. All energy figures are analytic bounds from the BitNet literature, not measurements of our system.

We state these plainly because the paper’s value is the framework: even if every number above moves under measurement, the bit-budget equation, the optimal-threshold result, and the router-leverage argument stand as stated.

11 Conclusion

1.58 bits was never a wall; it was a waypoint. Below it lies an analytic theory of when the zero state earns its keep (about 21% error reduction at the optimum, under a Laplace prior). Around it lies a budgeting problem with a water-filling solution that retires uniform precision. Above it—in the mixture-of-experts, where precision has the most leverage—lies a fully-ternary council that spends its bits on *decisions* rather than *parameters*. One equation governs all three. The floor, it turns out, was a design choice all along.

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